THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Thermosphere	Ionosphere
2	**(a)** Both A and R are true and R is the correct explanation of A.	Moisture-laden
3	**(c)** land breeze	Land to sea
4	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but different layers
5	**(d)** A is false but R is true.	A false, R true
6	**(c)** thermosphere	Ionosphere
7	**(a)** Both A and R are true and R is the correct explanation of A.	Carbon sink
8	**(a)** Both A and R are true and R is the correct explanation of A.	Crossing the sea
9	**(b)** Using solar energy	Clean energy
10	**(b)** land warmer than sea	Day

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorbs X-rays and short UV (1); Few molecules absorb high energy (1).	1+1
12	Rainfall may be concentrated (1); Poor drainage and silt (1).	1+1
13	Better forecasts (1); Helps plan crops and water (1).	1+1
14	Heat trapped by gases (1); Poor air quality hurts health (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Roots hold soil (1); Reduce flow speed (1); Trap silt (1).	1+1+1
16	Stratosphere has no weather (1); Forecasts need troposphere data (1); Cloud and rain occur there (1).	1+1+1
17	Rain varies by place (1); Local drainage and rivers (1); Better decisions (1).	1+1+1
	OR: Sowing is delayed (1); Crop yield falls (1); Food prices and rural incomes affected (1).	1+1+1
18	IMD: winter, summer, monsoon, post-monsoon (1); Rtus are six, two months each (1); Both reflect the annual cycle (1).	1+1+1
	OR: Cool sea air at day (1); Fishermen use land breeze at night (1); Comfortable climate for tourists (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heavy rain and western disturbances (1); Rivers already high (1); Weak embankments (1); Construction near rivers and silt (1); Repair, desilting, early warnings (1).	1+1+1+1+1
	OR: Agriculture depends on rain (1); Reservoirs and rivers fill (1); Rural income and food prices (1); Floods and drought risks (1); Irrigation and storage reduce dependence (1).	1+1+1+1+1
20	Greenhouse gases rise (1); Warming, floods, droughts (1); Impact on health and farming (1); Reduce carbon footprint (1); Collective policies (1).	1+1+1+1+1
	OR: Different time scales (1); Weather affects daily activity (1); Climate affects crops and housing (1); Monsoon as example (1); Forecast and planning (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** light breeze	6-11 km/hr
2	**(a)** 17 September	Begin of withdrawal
3	**(b)** 22 May	Starts Kerala
4	**(a)** global warming	Heat trapped
5	**(a)** Both A and R are true and R is the correct explanation of A.	Human factors
6	**(a)** Both A and R are true and R is the correct explanation of A.	Differential cooling
7	**(d)** very high impact	Scale 1-4
8	**(c)** Frequent use of petrol car	Fossil fuels
9	**(b)** Altitude brings autumn and spring	Local climates
10	**(c)** the monsoon is weak	Low rain

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Predict rain timing (1); Plan sowing and harvest (1).	1+1
12	Sea moderates temperature (1); Breezes cool by day and warm by night (1).	1+1
13	Shows where emissions arise (1); Helps plan reduction (1).	1+1
14	Weak old embankments (1); Houses built close to rivers (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat stress and disease (1); Floods and drought damage crops (1); Livelihoods suffer (1).	1+1+1
16	Rain pattern follows seasons (1); Planting is timed by season (1); Tradition and science combine (1).	1+1+1
17	Forecasts have limits (1); People must act on warnings (1); NDMA guidelines help (1).	1+1+1
	OR: Air thinner at height (1); Less weight above (1); Pressure falls rapidly with altitude (1).	1+1+1
18	Gases trap heat (1); Aerosols reflect sunlight (1); Change temperature and rain patterns (1).	1+1+1
	OR: Heavy rain upstream (1); Weak embankments and settlements near river (1); Low-lying lands flooded (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Differential heating of land and sea (1); Low pressure inland draws moist winds (1); SW monsoon rain June-September (1); NE monsoon rain to east coast (1); Agriculture, water and economy depend on it (1).	1+1+1+1+1
	OR: Pressure falls with height (1); Temperature changes by layer (1); Oxygen thins (1); Mountaineers need oxygen (1); Ozone protects life (1).	1+1+1+1+1
20	Insolation drives temperature (1); Warm air rises, low pressure (1); Winds carry moisture (1); Precipitation follows (1); Averages over 30 years form climate (1).	1+1+1+1+1
	OR: Krishiparashara and Brihatsamhita (1); Nakshatra observation (1); Kalidasa's monsoon account (1); NMM and Mission Mausam (1); Joint use helps farmers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** A is false but R is true.	A false: different time scale
2	**(c)** Troposphere, stratosphere, mesosphere, thermosphere, exosphere	Order from the surface
3	**(c)** climate	Long-term pattern
4	**(b)** Solar panels	Renewable energy
5	**(a)** Both A and R are true and R is the correct explanation of A.	Weather-free
6	**(d)** divide the Moon's path into 27 parts	Traditional astronomy
7	**(b)** Insolation decreases	Sun angle
8	**(b)** internal pressure equals external	Balanced
9	**(a)** Both A and R are true and R is the correct explanation of A.	Differential heating
10	**(c)** Strengthening embankments and desilting	Flood management

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorb heat slowly (1); Water volume rises with melting ice (1).	1+1
12	Weather varies daily (1); Climate needs 30-year data (1).	1+1
13	Preparedness for climate change (1); Using better forecasts (1).	1+1
14	Traps heat (1); Shields from radiation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Clean water and sanitation (1); Drain standing water (1); Medical camps (1).	1+1+1
16	Fossil fuels add CO2 (1); Forests absorb CO2 (1); Net balance decides footprint (1).	1+1+1
17	Warm humid air feels sticky (1); Sweat does not evaporate (1); Coastal cities in summer (1).	1+1+1
	OR: Over land it is dry (1); Over Bay of Bengal it picks up moisture (1); Then rains on east coast (1).	1+1+1
18	Save electricity and water (1); Encourage walking, cycling (1); Reduce plastic and plant trees (1).	1+1+1
	OR: Strong heating at equator lifts air (1); Poles are colder (1); Higher tropopause at equator (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Monsoon direction (1); Distance from sea (1); Mountains (1); NE monsoon on east coast (1); Local winds and pressure (1).	1+1+1+1+1
	OR: Heavier rain from warming (1); Settlements on floodplains (1); Poor drainage and embankments (1); Loss of life and property (1); Adaptation and preparedness (1).	1+1+1+1+1
20	Troposphere for air and weather (1); Stratosphere's ozone filters UV (1); Mesosphere burns meteors (1); Thermosphere reflects radio waves (1); Gravity holds the whole blanket (1).	1+1+1+1+1
	OR: Personal steps reduce footprint (1); Governments set policy (1); Industries must cut emissions (1); Collective action is needed (1); Awareness spreads change (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.